Z1 jog pendant – parts list and build guide

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Overview

A handheld jog pendant for the Makera Z1 Pro. It has a 60 mm MPG handwheel, two 8-position rotary switches (axis and step) and a 0.96″ OLED that shows the coordinate of the selected axis. The enclosure is 3D printed and measures 85 × 178 × 25.5 mm including the cable ear.

A Pro Micro (ATmega32U4) emulates the commercial **WHB04B** USB pendant (USB ID 10CE:EB93). Carvera Community Controller therefore detects it with no driver or software change.

- The pendant sends axis, step and handwheel clicks to the tablet every 10 ms.
- The Controller turns that into jog moves on the Z1 and sends coordinates back for the display.
- In the OFF position the display is blank and the wheel does nothing.
- A mode switch selects an alternative mode for the NestWorks C500. This mode needs a bridge program on the tablet that does not exist yet.

Parts list

#	Part	Qty	Specification	Example source
1	Pro Micro	1	ATmega32U4, **5 V / 16 MHz** (not the 3.3 V / 8 MHz version), micro-USB or USB-C	Hadex M381B
2	OLED display 0.96″	1	128 × 64, SSD1306 controller, I2C, 4 pins (GND, VCC, SCL, SDA)	Hadex M508
3	Rotary switch RS16	2	1 pole × 8 positions, D shaft Ø6 mm (flat 1.5 mm), M9 thread, panel hole 8.8 mm, body Ø16 mm	L333F

#	Part	Qty	Specification	Example source
4	MPG handwheel	1	Ø60 mm, 100 pulses per revolution, 5 V supply, A/B outputs, 3 × M3 rear studs on 53 mm circle	AliExpress / eBay (“CNC MPG 60 mm 5 V”)
5	USB cable	1	4 cores, outer Ø ⩽ 4 mm, plug to suit the tablet (USB-A or USB-C)	any
6	Display cover glass	1	26.9 × 19.5 mm, max. 1 mm thick, cut from a microscope slide	any
7	Enclosure screws	7	self-tapping 3 × 10 mm, **or** M3 × 8 (insert variant), **or** M3 × 10 countersunk (nut variant)	any
8	M3 heat-set inserts	7	insert variant only; hole Ø4 × 6 mm	Hadex P131L / M / N
9	M3 nuts	7	nut variant only; placed during a print pause at 19.5 mm	any
10	M3 nuts for the MPG	3	on the handwheel studs	any
11	Hook-up wire	approx. 1 m	0.14–0.25 mm² stranded, several colours	any
12	ON-OFF toggle switch	1	optional, Z1 / C500 mode	any
13	Heat-shrink tubing	some	Ø1.5–2 mm	any

All inputs use the Pro Micro’s internal pull-ups, so no resistors are needed.

Tools and consumables

- 3D printer with 0.4 mm nozzle. A multi-material unit is optional, for the two-colour label insert.
- Filament: PETG black for the body, PETG or PLA white for the labels.
- Soldering iron (fine tip), solder, flux.

- Wire stripper, side cutters, tweezers.
- Soldering iron tip for heat-set inserts (insert variant only).
- Glass cutter or diamond scribe for the cover glass.
- CA glue (thin) for the glass and white inlays.
- Multimeter, to check the switch positions and shorts.
- Windows PC with a USB data cable for flashing (charging-only cables do not work).

Printed parts and print settings

File	Qty	Material	Note
teleso_vymenne_popisky_samorezne.stl	1	PETG black	body for self-tapping screws
teleso_vymenne_popisky_lisovane_vlozky.stl	1	PETG black	body for heat-set inserts
teleso_vymenne_popisky_matice.stl	1	PETG black	body for nuts; pause at 19.5 mm and drop in 7 M3 nuts
teleso_popisky_multimaterial_*.3mf	1	PETG black + white	body with labels printed in (MMU)
dno.stl	1	PETG black	bottom plate with Pro Micro holder and cable channel
vlozka_zvysena_MMU.3mf	1	black + white	raised label insert, two colours, already face down
vlozka_popisky_zahloubene.stl	1	PETG black	single-colour label insert
vypln_popisky_bila.stl	1	white	inlays for the single-colour insert
knoflik.stl	2	PETG black	knobs with D bore
vypln_ryska_knofliku_2ks_bila.stl	1	white	pointer inlays for both knobs

Print only **one** body variant and **one** label insert variant.

Settings:

- Nozzle 0.4 mm, layer height 0.16 mm.
- No supports. All overhangs are 45° or less.
- Body face down on a textured plate, for a nicer front surface.
- Bottom plate flat side down.
- Turn on elephant-foot compensation in the slicer, otherwise the inlays will not fit.
- Infill 20–30 %, 3 walls.

Wiring

Pro Micro pin	Connects to	Note
VCC	MPG Vcc (+5 V), OLED VCC	5 V from USB
GND	MPG 0V, OLED GND, common (centre) terminal of both RS16 switches, one leg of the mode switch	one common ground
RX1 (D0)	MPG A	interrupt input
TX0 (D1)	MPG B	interrupt input
2 (SDA)	OLED SDA	I2C
3 (SCL)	OLED SCL	I2C
4	axis switch, position 1 = OFF	
5	axis switch, position 2 = X	
6	axis switch, position 3 = Y	
7	axis switch, position 4 = Z	
8	axis switch, position 5 = A	positions 6–8 not connected
9	step switch, position 1 = 0.001 mm	
10	step switch, position 2 = 0.01 mm	
16	step switch, position 3 = 0.1 mm	
14	step switch, position 4 = 1 mm	positions 5–8 not connected
15	mode switch, other leg	closed to GND = C500 mode
USB	cable to the tablet	VBUS red, D− white, D+ green, GND black

Notes:

- **MPG:** wire only the first terminal block (A, B, 0V, Vcc). The second block (A−, B−, inverted signals) stays unconnected.
- **RS16:** the common terminal is in the centre. Turn the switch fully anticlockwise, then find position 1 with a multimeter (continuity between the centre and that terminal). Count the following positions clockwise.
- Pins A0–A3 (18–21) and RAW stay free.

- **USB cable:** solder it to the Pro Micro's USB pads, or cut a short micro-USB / USB-C plug and plug it in. If the colours of your cable differ, check with a multimeter.

Assembly

Tip: flash and test the firmware (next section) on the bare Pro Micro before you solder it in, so a bad board is found early.

1. **Clean the prints.** Remove stringing from the holes and check that the knobs slide onto the RS16 D shafts.
2. **Inserts (insert variant).** Press the 7 M3 heat-set inserts into the body with a soldering iron at about 230 °C. Keep them square.
3. **Cover glass.** Cut the glass to 26.9 × 19.5 mm and glue it into the display window from the inside with a thin bead of CA glue on the edge only.
4. **Label insert.** Single-colour version: press the white inlays in first, with a drop of CA glue if loose. Then put the insert into the body face.
5. **RS16 switches.** Push each switch through from inside, with the anti-rotation tab in its slot. Tighten the nut from the front; the washer can be left out.
6. **MPG.** Insert the handwheel from the front and fix it with 3 M3 nuts on the studs from inside.
7. **OLED.** Snap the display into its bay with the pins pointing down, toward the switches.
8. **Wires to the parts.** Solder the wires to the switches, MPG and OLED according to the wiring table. Leave about 5 cm of slack, so the bottom plate can be opened. Put heat-shrink over the joints.
9. **Pro Micro.** Clip the board into the holder on the bottom plate, under the handwheel, with the USB connector toward the ear. Solder the wires to it.
10. **USB cable.** Lay the cable in the channel of the bottom plate and lead it out through the ear. Tie a knot or cable tie inside as strain relief.
11. **Check.** Before closing, measure with a multimeter that VCC and GND are not shorted.
12. **Close.** Fit the bottom plate with 7 screws. Turn both switches fully anticlockwise and push the knobs on with the pointer at the first label (OFF and 0.001).

Compiling and flashing the firmware

The firmware package (jog_ovladac_firmware.zip) contains:

- **whb04_pendant/whb04_pendant.ino**: the sketch. The folder and the file must keep the same name.
- **boards.local.txt**: sets the USB identity of the pendant.

- **99-whb04-pendant.rules**: udev rule, only for a Linux tablet.
- **ZAPOJENI.md**: wiring notes in Czech.

1. Install Arduino IDE 2

1. Download Arduino IDE 2 for Windows from arduino.cc/en/software and install it with the default options. Allow the USB driver installation when Windows asks.
2. Start the IDE once and close it again. This creates the folder %LOCALAPPDATA%\Arduino15.

2. Install the AVR board package

1. Open **Tools → Board → Boards Manager**.
2. Search for **Arduino AVR Boards** and install it. Version 1.8.8 was tested.

3. Install the U8g2 library

1. Open **Tools → Manage Libraries**.
2. Search for **U8g2** (by olikraus) and click **Install**. If it asks about dependencies, choose **Install all**.

HID.h and Wire.h come with the AVR package and need no install.

4. Copy boards.local.txt

This file makes every sketch built for “Arduino Leonardo” report itself as USB ID **10CE:EB93**, the ID of the WHB04B pendant that Carvera Controller looks for.

1. Close the Arduino IDE.
2. Press **Win + R**, type `%LOCALAPPDATA%\Arduino15\packages\arduino\hardware\avr` and press Enter.
3. Open the folder with the version number, for example **1.8.8**. It already contains **boards.txt**.
4. Copy **boards.local.txt** into this folder, next to boards.txt.
5. Start the IDE again.

The file contains:

```
leonardo.build.vid=0x10CE
leonardo.build.pid=0xEB93
leonardo.build.usb_product="MPG pendant (WHB04B kompat.)"
leonardo.build.usb_manufacturer="DIY"
```

The setting applies to every sketch uploaded as Leonardo. For other Leonardo projects, delete the file and restart the IDE. If you update the AVR package to a new version, copy the file into the new version folder again.

5. Open the sketch and check the settings

1. **File → Open** and select **whb04_pendant.ino**.
2. At the top of the sketch you can change:
   - `ENC_COUNTS_PER_DETENT` (default 4): signal changes per click of the wheel. If one click moves by 2 or 4 steps, or every second click does nothing, change it to 2 or 1.
   - `ENC_REVERSE` (default false): set true if the wheel moves the wrong way.
   - `DISPLAY_FLIP` (default false): set true if the display image is upside down.

6. Select board and port

1. Connect the Pro Micro to the PC with a USB data cable.
2. **Tools → Board → Arduino AVR Boards → Arduino Leonardo**.
3. **Tools → Port** and select the COM port of the board. If you are not sure, unplug the board, look at the list, plug it back in and take the port that appeared.

After the first upload the board no longer reports itself as a Leonardo, so the IDE may show the port without a board name. That is fine.

7. Compile

1. Click **Verify** (✓, top left).
2. Wait for **Done compiling** at the bottom. The output shows memory usage: about 25 000 bytes (86–90 %) of program memory and about 1 200 bytes of RAM.
3. If you get an error about U8g2lib.h, the library is missing (step 3). If you get “Compiled sketch not found”, you clicked something other than Verify; click Verify again.

8. Upload

1. Click **Upload** (→). The IDE compiles again, resets the board and uploads.
2. Wait for **Done uploading**.

If the upload stops at “Waiting for upload port” or the port disappears:

1. Click Upload again.
2. As soon as the bottom shows **Uploading…**, short the **RST** pin to **GND** twice quickly with a piece of wire or tweezers.
3. The board stays in the bootloader for about 8 seconds, and the IDE finds it.

Warning: do not select a 3.3 V / 8 MHz board (such as SparkFun Pro Micro 3.3 V) for a 5 V board. The USB will then stop working until you upload correctly with the double-reset method.

9. Alternative: arduino-cli

If you prefer the command line (with arduino-cli installed and the AVR package and U8g2 installed as above):

```
arduino-cli compile --fqbn arduino:avr:leonardo whb04_pendant
arduino-cli upload  --fqbn arduino:avr:leonardo -p COM5 whb04_pendant
```

Replace COM5 with your port.

10. Check the USB identity

1. Open **Device Manager** (right-click Start → Device Manager).
2. Under **Human Interface Devices**, open the new HID device → **Details** → **Hardware Ids**.
3. It must show **VID_10CE&PID_EB93**, with two entries **Col01** and **Col02**. If you see VID_2341 instead, boards.local.txt was not used: check its folder (step 4) and upload again.

First test and troubleshooting

1. Plug the pendant into the tablet with the mode switch open (Z1 mode).
2. Start Carvera Community Controller and connect to the Z1.
3. Turn the axis switch from OFF to X. The display shows the X coordinate.
4. Select step 0.1 and turn the wheel one click. X must move by 0.1 mm.
5. Test every axis and every step. In OFF the display goes blank and the wheel does nothing.

Problem	Cause and fix
Controller does not see the pendant	Check the USB ID in Device Manager (VID_10CE&PID_EB93). Try another cable or port.
Wheel moves the wrong way	Set `ENC_REVERSE = true` and upload again, or swap A and B.
One click moves several steps, or every other click does nothing	Change `ENC_COUNTS_PER_DETENT` to 2 or 1.

Problem	Cause and fix
Wheel does nothing	Check Vcc and 0V on the MPG, and that A/B go to D0/D1.
Wrong axis or step selected	Check the switch position wiring with a multimeter; position 1 must be at the fully anticlockwise end.
Display stays dark	Check SDA/SCL (not swapped) and VCC. The display is also dark in OFF and before the Controller sends coordinates.
Display upside down	Set `DISPLAY_FLIP = true`.
Display shows REF	Only in C500 mode: the machine is not homed.

Linux tablet

On Linux, normal users cannot open the pendant without a udev rule:

1. Copy the rule file: `sudo cp 99-whb04-pendant.rules /etc/udev/rules.d/`
2. Reload the rules: `sudo udevadm control --reload && sudo udevadm trigger`
3. Unplug the pendant and plug it in again.

Open points

- The hole for the mode switch is not in the enclosure model yet. It needs a position and the thread diameter of the chosen switch.
- C500 mode needs a bridge program on the tablet, because the C500 communicates over gRPC. This needs the .proto definitions from Nest Studio.